FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. Use $\int_0^4 F dt = mv - m(5)$

$\text{work}_F = \frac{1}{2}mv^2 - \frac{1}{2}m(5)^2$

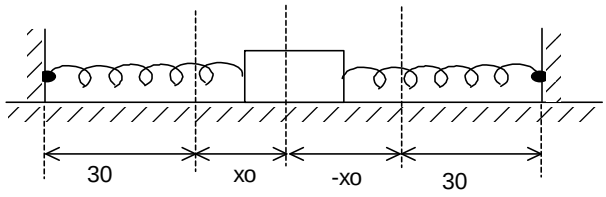
2. A

Sol. For equilibrium of the block

$K_1(x_o + 30 - 20) = K_2[30 - x_o - 20]$

$\Rightarrow x_o = 5$ cm

$T = 2\pi\sqrt{\frac{m}{K_i + K_2}} = \frac{\pi}{100}$ sec



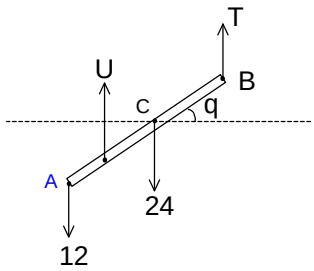
3. C

Sol. U = up thrust = $A\left(\frac{\ell}{2}\right)P_w g$

T + U = 12 g + 24 g

Taking torque about B,

$U\left(\frac{\ell}{2} + \frac{\ell}{4}\right)\cos\theta = 12g(\ell\cos\theta) + 24g\left(\frac{\ell}{2}\cos\theta\right)$



4. D

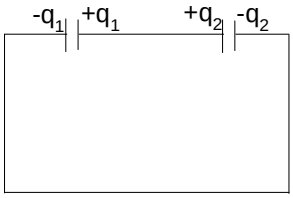
Sol. After a very long time inductor behaves like zero resistance.

5. B

Sol. $\because \lambda_1 N_1 = \lambda_2 N_2 \Rightarrow \frac{N_{Ra}}{N_{Rn}} = \frac{\lambda_{Rn}}{\lambda_{Ra}} = \frac{t_{Ra}^{\frac{1}{2}}}{t_{Rn}^{\frac{1}{2}}}$

6. B

Sol. Let q_1 and q_2 be the charge on each capacitor when current its maximum.



$q_1 + q_2 = 10CV_0 \quad .....(i)$

$\frac{q_1}{2c} = \frac{q_2}{4C}$

$\Rightarrow q_2 = 2q_1$

From (i) and (ii), $q_1 = \frac{10CV_0}{3} \quad .....(ii)$

$\therefore$ PD across each capacitor is $\frac{5V_0}{3}$

From conservation of energy.

$\frac{1}{2}(2C)V_0^2 + \frac{1}{2}(4C)(9V_0^2) = \frac{1}{2}(6C)\left(\frac{5V_0}{3}\right)^3 + \frac{1}{2}Li^2$

$\therefore i = 8V_0\sqrt{\frac{C}{3L}}$

7. B

Sol. $h = u\cos\theta t - \frac{1}{2}gt^2$

This is a quadratic equation in t, which will have two roots – say t_1 & t_2

$\therefore t_1 + t_2 = \frac{2u\cos\theta}{g} \Rightarrow \frac{2u\cos\theta}{g} = 4$

& $t_1 t_2 = \frac{2h}{g}$

maximum height = $\frac{(u\cos\theta)^2}{2g}$

8. BCD

Sol. $\overrightarrow{v_B} = 8\hat{j} + 2t\hat{k}$

$\overrightarrow{v_C} = \overrightarrow{v_o} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$

$\overrightarrow{r_B} = 8t\hat{j} + t^2\hat{k}$

$\overrightarrow{r_C} = v_x t\hat{i} + v_y t\hat{j} + v_z t\hat{k}$

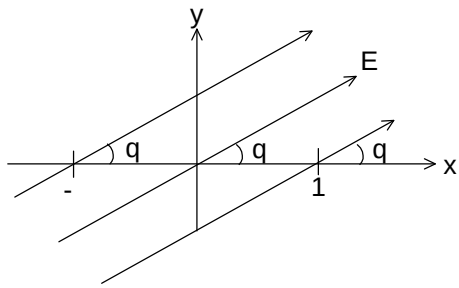
At 4 sec, $\overrightarrow{r_B} = \overrightarrow{r_C} \Rightarrow v_x = 0, v_y = 8m/s$ & $v_z = 4m/s$

$\therefore \overrightarrow{v_o} = \left(8\hat{j} + 4\hat{k}\right) m/s$

9. BCD

Sol. $(E\cos\theta)2 = 20$, where $0 \le \theta < 90^o$

$\Rightarrow E\cos\theta = 10$



10. ACD

Sol. When a compression pulse strikes a fixed surface, it reflects as compression pulse.

11. BC

Sol. Use $\Delta x = n\lambda$ (for maxima)

& $\Delta x = \left(n + \frac{1}{2}\right)\lambda$ (for minima)

12. AC

Sol. $\vec{E} = 54\hat{i} + 72\hat{j}$, $E = 90 NC^{-1}$

$90 = \frac{9\times10^9 Q}{r^2}$ ....(i)

$V = 1800 = \frac{9\times10^9 Q}{r}$ .....(ii)

From equations (i) and (ii)

r = 20 m, q = 4 mC

Now , $\frac{(9-x_0)\hat{i} + (4-y_0)\hat{j}}{20} = \frac{54\hat{i} + 72\hat{j}}{90}$

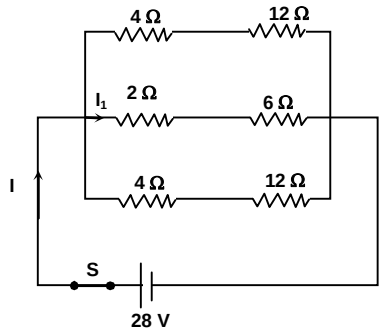
$\Rightarrow x_0 = -3, y_0 = -12$

13. BD

Sol. $V = \int_2^4 \left(\frac{25t - 50}{10}\right) dt$

14. BC

Sol. Equivalent circuit is

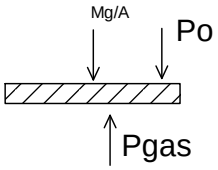


15. B

Sol. $Q = nC_P\Delta T = C_P\left(\frac{P\Delta V}{R}\right)$

16. A

Sol.



17. B

Sol. Apply conservation of energy and conservation of angular momentum.

18. B

Sol. Area velocity $= \frac{\text{Angular momentum}}{2m}$.

Chemistry

PART – II

SECTION – A

19. B

Sol. $H^+ + OH^- \longrightarrow H_2O(l)$

$\left(\Delta H_{neut}\right)_{SA+SB} = \left(\Delta H_{neut}\right)_{H^+ + OH^-} = -285 + 227.3 = -57.7 \text{ kJ}$

$-55 = \left(\Delta H_{ion}\right)_{CH3COOH} - 57.7$

$\left(\Delta H_{ion}\right)_{CH3COOH} = 2.7 \text{ kJ/mol}$

20. D

Sol. Zn^{+2} forms soluble complex with excess KCN but does not react with excess KI

Pb^{+2} is soluble in excess KI but insoluble in excess KCN

Ag^+ is soluble in excess KCN but insoluble in excess KI

Hg_2^{+2} gives ppt of Hg when reacts with KI as well as well as excess KCN

$Hg_2^{2+} + 2I^- \longrightarrow Hg_2I_2 \downarrow$

$Hg_2I_2 \downarrow + 2I^- \longrightarrow [HgI_4]^{2-} + Hg \downarrow$

$Hg_2^{2+} + 2CN^- \longrightarrow \underset{\text{(soluble)}}{Hg(CN)_2} + Hg \downarrow$

21. C

Sol.

	C_2H_2	+ CO +	O_2	$\longrightarrow$ CO_2 + H_2O
Volume initially	x ml	(20 – x) ml	30 ml	
Volume left			8 ml	26 ml

2 × Volume of C_2H_2 taken + volume of CO taken = Volume of CO_2.

2 x + 20 – x = 26 ; x = 6 ml.

Volume of C_2H_2 taken = 6 ml and volume of CO taken = 14 ml.

22. B

Sol. Cell reaction is:

$Ag(s)/AgBr(s)/Br^-(aq)\,||\,Cl^-(aq)/AgCl(s)/Ag(s)$

$Ag_c^+ \longrightarrow Ag_a^+$

$E_{cell} = 0 - 0.059\log\left(\frac{\left[Ag^+\right]_a}{\left[Ag^+\right]_c}\right)$

$\Rightarrow 0 = 0 - 0.059\log\left(\frac{\left[Ag^+\right]_a}{\left[Ag^+\right]_c}\right)$

$\therefore \left[Ag^+\right]_a = \left[Ag^+\right]_c \Rightarrow \frac{K_{sp} \text{ of AgBr}}{\left[Br^-\right]} = \frac{K_{sp} \text{ of AgCl}}{\left[Cl^-\right]}$

or, $\frac{8\times10^{-13}}{10^{-10}} = \frac{\left[Br^-\right]}{\left[Cl^-\right]}$

23. D

Sol. Tertiary amine does not form H-bond among themselves because H-atoms are not directly joined with N- atom

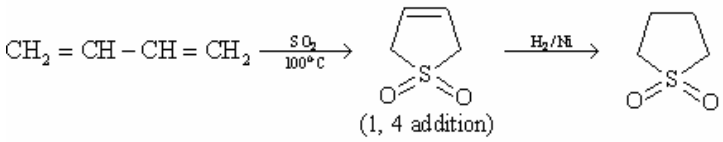
24. D

Sol. For a Bcc lattice, $\frac{X^+}{Y^-} = 0.732 \Rightarrow Y^- = \frac{0.4}{0.732} = 0.546^0\,A$

Now for fcc lattice, $\frac{Z^+}{Y^-} = 0.414 \Rightarrow Z^+ = 0.546 \times 0.414 = 0.226^0\,A$

25. B

Sol.



26. B

Sol. Theory based

27. ABCD

Sol. The atom directly joined with phenyl contains lone pair of electron and hence o-& p-director

28. ABC

Sol. (A) is Hoffman degradation
(C) is NGP reaction and hence retention takes place
(D) is Gatterman aldehyde synthesis

29. ACD

Sol. Theory based

30. BC

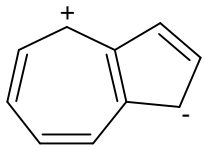
Sol. For Q using steady state concept
Rate of appearance of Q = rate of disappearance of Q
$K_1[P] = K_2[Q]$

Number of nuclei of Q = $\frac{k_1}{k_2} \times$ no. of nuclei of P $= \frac{10/60}{1000} \times 6 \times 10^{23} = 10^{20}$

For R, since rate constant is very high for second step than first step
Number of nuclei of R = Number of nuclei of P disintegrated = 6×10^{23}

31. ABC

Sol.



32. AD

Sol. New compound is brown ring, $[Fe(H_2O)_5(NO)]SO_4$ where NO has +1 charge. Due to NO^{+1}, only $4s^1$ electron is paired up in 3d subshell.

33. B

Sol. $\underset{(A)}{FeSO_4.7H_2O} \xrightarrow{\Delta} \underset{(B)}{Fe_2O_3}(s) + SO_2(g) + SO_3(g) + H_2O\ (vap.)$

34. D

Sol. F is $CuSO_4.5H_2O$

35. B

Sol. $\underset{(S)}{CH_3-CH_2-COOH} \xrightarrow{N_3H,H^+/\Delta} \underset{(P)}{CH_3-CH_2-NH_2} \xleftarrow{Br_2/KOH} \underset{(Q)}{CH_3-CH_2-CONH_2} \xrightarrow{LiAlH_4} \underset{(R)}{CH_3-CH_2-CH_2-NH_2}$

36. D

Sol. $\underset{(S)}{CH_3-CH_2-COOH} \xrightarrow{N_3H,H^+/\Delta} \underset{(P)}{CH_3-CH_2-NH_2} \xleftarrow{Br_2/KOH} \underset{(Q)}{CH_3-CH_2-CONH_2} \xrightarrow{LiAlH_4} \underset{(R)}{CH_3-CH_2-CH_2-NH_2}$

The conversion of (S) to (P) is Schmidt reaction.

Mathematics

PART – III

SECTION – A

37. A

Sol. From the figure,

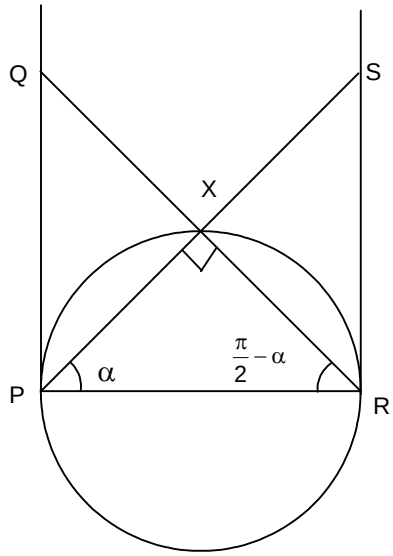
$$\tan\alpha = \frac{RS}{PR} = \frac{RS}{2r} \quad \ldots.(i)$$

and $\tan\left(\frac{\pi}{2}-\alpha\right) = \frac{PQ}{PR} = \frac{PQ}{2r}$ ....(ii)

Multiplying (i) and (ii), we get

$$\tan\alpha\cot\alpha = \frac{RS.\,PQ}{(2r)^2}$$

$$\therefore \quad 2r = \sqrt{(RS)(PQ)}$$



38. D

Sol. For ${}^{10}C_x$ to be defined $x \in I, 0 \le x \le 10$.

$$\frac{x}{{}^{10}C_x} \le 1 \Rightarrow x \ne 10 \qquad [x] = x, \{x\} = 0 \text{ as } x \in I$$

$$\Rightarrow 5x^2 - 60x + 160 > 0$$

$$\Rightarrow x^2 - 12x + 32 > 0$$

$$\Rightarrow x \notin [4,8]$$

39. B

Sol.

$$n(E) = \frac{1}{2}\left[{}^4C_1.{}^2C_1 + {}^{24}C_1.{}^3C_1 + {}^{36}C_1.{}^4C_1\right] = 112$$

$$\Rightarrow P(E) = \frac{112}{{}^{64}C_2} = \frac{1}{18}$$

40. D

Sol. Clearly incentre of ΔDEF is orthocenter of ΔABC is i.e (α,α) which lies on hyperbola

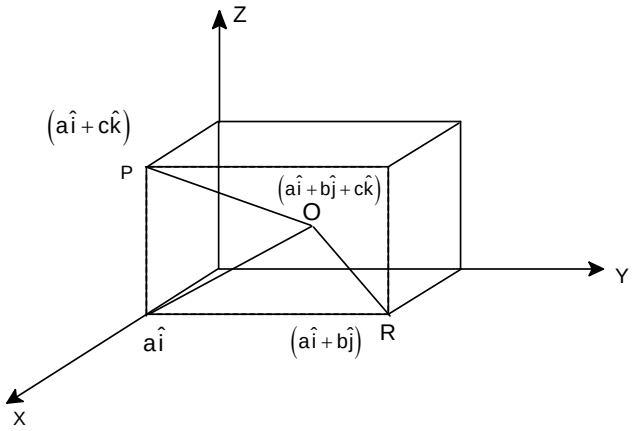
$xy - x - y - 3 = 0 \Rightarrow \alpha = 3$

Again equation of DE is $x + y - \frac{26}{5} = 0$

$\Rightarrow r =$ length of perpendicular from $I(3,3)$ on DE $\Rightarrow r = \frac{2}{5}\sqrt{2}$ units

41. C

Sol.



42. A

Sol. Take m = [x]

Observe that $\sqrt{m^2+2(n+1)^2}-\sqrt{m^2+2n^2}=\dfrac{4n+2}{\sqrt{m^2+2(n+1)^2}+\sqrt{m^2+2n^2}}$

Which is less than $\dfrac{2n+1}{m}$.Taking

$m \le \sqrt{m^2 2n^2} < m+1$ or $0 \le n < \sqrt{m}$

We get $\sqrt{m^2+2(n+1)^2}-\sqrt{m^2+2n^2} < \dfrac{2}{\sqrt{m}}+\dfrac{1}{m} \le \dfrac{3}{\sqrt{m}}$

Hence we conclude that $0 < \left|x-\sqrt{m^2+2n^2}\right| < \dfrac{3}{\sqrt{m}}$

Applying squeeze play rule $\lim\limits_{x\to\infty} f(x) = \lim\limits_{x\to\infty} f(x) = 0$

43. C

Sol. Differentiating both sides w.r.t b, we get

$$\int_0^{\frac{\pi}{2}} \frac{-\cos x}{(a+b\cos x)^2}dx = \frac{1}{b^2-a^2}-\frac{b}{(b^2-a^2)^{\frac{3}{2}}}\ln\left(\frac{b+\sqrt{b^2-a^2}}{a}\right)$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{\cos x}{(3+5\cos x)^2}dx = -\frac{1}{16}+\frac{5}{64}\ln 3$$

44. A, C

Sol. $k_1 = \tan 27\theta - \tan\theta = (\tan 27\theta - \tan 9\theta)+(\tan 9\theta - \tan 3\theta)+(\tan 3\theta - \tan\theta)$

Now, $\tan 3\theta - \tan\theta = \dfrac{\sin 2\theta}{\cos 3\theta \cos\theta} = \dfrac{2\sin\theta}{\cos 3\theta}$

Similarly, $\tan 9\theta - \tan 3\theta = \dfrac{2\sin 3\theta}{\cos 9\theta}$

and $\tan 27\theta - \tan 9\theta = \dfrac{2\sin 9\theta}{\cos 27\theta}$

$$\therefore\ k_1 = 2\left(\frac{\sin 9\theta}{\cos 27\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin\theta}{\cos 3\theta}\right) = 2k_2$$

45. A, C, D

Sol. $f(x+y) = 2^x f(y) + 4^y f(x)$ .....(1)

replace x into y and x, we get

$f(x+y) = 2^y f(x) + 4^x f(y)$

$\Rightarrow 2^x f(y) + 4^y f(x) = 2^y f(x) + 4^x f(y).....(2)$

$\Rightarrow \frac{f(x)}{4^x - 2^x} = \frac{f(y)}{4^y - 2^y} = k$

$\Rightarrow f(x) = k\left(4^x - 2^x\right)$

since, $f'(0) = \ln 2$, we get $k = 1$.

Hence, $f(x) = 4^x - 2^x$.

46. A, D

Sol. Let O be the intersection of diagonals AC and BD.

Let position vector of A $(\vec{a}), B(\vec{b}), C(\vec{c}), D(\vec{d})$. Let position vector of G be $(\vec{g})$

Let $\frac{AM}{AB} = m, \frac{BN}{BC} = n, \frac{DP}{DC} = p$

Now, $\overrightarrow{OM} = (1-m)\vec{a} + m\vec{b}$

$\overrightarrow{OP} = (1-p)\vec{d} + p\vec{c}$

$\Rightarrow \vec{g} = \frac{(1-m)\vec{a} + n\vec{b} + (1-n)\vec{b} + n\vec{c} + (1-p)\vec{d} + p\vec{c}}{3}$ as $\vec{a} = -\vec{c}$ and $\vec{d} = -\vec{b}$

$\vec{g} = (m+n+p-1)\vec{c} + (m-n+p)\vec{b}$

If $G(\vec{g})$ is collinear with AC

$\Rightarrow m - n + p = 0$

If $G(\vec{g})$ is collinear with BD

$\Rightarrow m + n + p - 1 = 0.$

47. A, B, C

Sol. The tangent $3x + 4y - 25 = 0$ is tangent at vertex and axis is $4x - 3y = 0$

So, PS = a = 5

Latus rectum = Ab = 20

48. A, C, D

Sol. $M^T M = I, M^T I = M^T$

$M^T M_1 = M_2$

$|M_1| = |M_2|$

49. A, B, C, D

Sol. $I = \int_0^{1/2}\left(\frac{1}{4} - x^2\right)^4 dx = \int_0^{1/2}\left(\frac{1}{4} - \left(\frac{1}{2} - x\right)^2\right)^4 dx = \int_0^{1/2}\left(x - x^2\right)^4 dx = \int_0^{1/2} x^4 (1-x)^4\, dx$

Also, by replacing x by (1 – x) we get, $I = \int_{1}^{1/2} (1-x)^4 x^4 dx = \int_{1/2}^{1} x^4 (1-x)^4 dx$

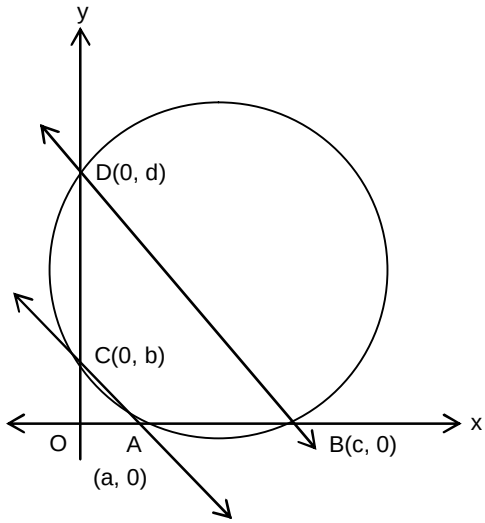
Adding we get, $2I = \int_{0}^{1} x^4 (1-x)^4 dx \Rightarrow I = \frac{1}{2}\int_{0}^{1} x^4 (1-x)^4 dx$. On putting $x = \sin^2\theta$ we get;

$I = \int_{0}^{\pi/2} \sin^9\theta \cos^9\theta d\theta = \frac{(8\times6\times4\times2)(8\times6\times4\times2)}{18\times16\times14\times12\times10\times8\times6\times4\times2} = \frac{1}{1260}$

50. A, B, C, D

Sol. If points A, B, C, D are concyclic, then ac = bd. The coordinates of the points of intersection of lines are $\left(\frac{ac(b-d)}{bc-ad}, \frac{bd(c-a)}{bc-ad}\right)$

Let coordinates of the point of intersection be (h, k)



Then, $h = \frac{ac(b-d)}{bc-ad}, k = \frac{bd(c-a)}{bc-ad}$

Given, $c^2 + a^2 = b^2 + d^2$. Since, ac= bd,

So, $(c-a)^2 = (b-d)^2$ or $(c-a) = \pm(b-d)$

Then the locus of the points of intersection is $y = \pm x$

51. C

Sol. Homogenizing, we get $3x^2 - y^2 - 2x(ax+by) + 4y(ax+by) = 0$. Since AB subtends a right angle at the origin, coefficient of x^2 + coefficient of y^2 = 0. This gives $(3-2a)+(-1+4b)=0$ or $a - 2b = 1$.

52. A, D

Sol. $a - 2b = 1$

$ax + by = 1$

Clearly, the line always passes through the point $(1,-2)$. Hence, $(h, k) = (1, -2)$.

53. D

Sol. $\cos 3n\alpha + i\sin 3n\alpha + \cos 3n\beta + i\sin 3n\beta + \cos n\gamma + i\sin 3n\gamma = 3$

$\left(\cos n\left(\alpha+\beta+\gamma\right) + i\sin\left(\alpha+\beta+\gamma\right)\right)$

54. C

Sol. $\cos 3n\alpha + i\sin 3n\alpha + \cos 3n\beta + i\sin 3n\beta + \cos n\gamma + i\sin 3n\gamma = 3$

$\left(\cos n\left(\alpha+\beta+\gamma\right) + i\sin\left(\alpha+\beta+\gamma\right)\right)$